7.4.1 Advantages of electrical transducers

The advantages of electrical transducers compared with other types of transducers are numerous. As to say some of the points

i. Electrical amplification is simple and easy.
ii. Frictional effects are minimized.
iii. Mass-inertia effects are minimized.
iv. Small power is used to control.
v. Transmission of electrical signal is easy.

The physical quantity is under measurement is first connected to primary transducers. This primary transducer changes the positions due to the external applied quantity. This variation is then connected to one of the arms of a Wheatstone bridge. Before applying the pressure, the bridge is brought to a balanced condition. The change in the resistance causes the bridge to unbalance. This deviation in the unbalance is measured in terms applied force. This is the basic principle of operation of transducers using wheatstone bridge method.

In a wheatstone bridge method, the four arms are connected as shown in figure 7.2 then one of the arm is connected to the unknown parameter to be find.

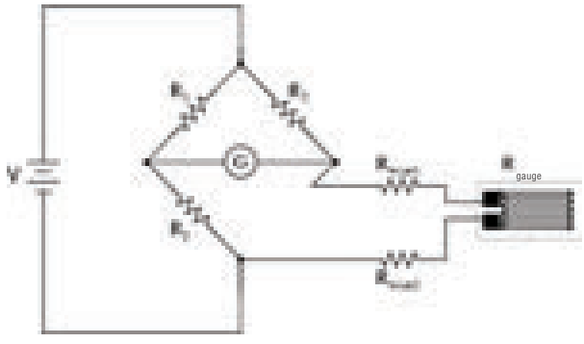


Fig. 7.2 Wheatstone bridge for strain gauge

7.5 Resistive, inductive and capacitive transducers

7.5.1 Resistive transducers

The types of transducers which work by changing its resistance are called resistive transducers. This type of transducer works both in AC and DC supply.

The resistance of metal conductor is given by $R = \frac{\rho l}{A}\ \Omega$

Where R = resistance

l = length of conductor

A = cross-sectional area of conductor

ρ = Resistivity of conductor material

Any method of varying the above said parameters by applying external force in the relationship is the very basic principle of operation for an electrical resistive transducer.

The Applications of resistive transducers are discussed below.

7.5.2 Measurement of displacement, pressure and force using resistive transducers

a. Potentiometers type transducers for linear displacement

A potentiometer is a three-terminal resistor with a sliding or rotating contact that forms an adjustable voltage divider. A resistive potentiometer or simply POT, consists of a resistive element provided with a sliding contact. Potentiometer transducers used to measure translational and rotational.

For example, let us consider the measurement of linear displacement

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